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22nd Asian Breast Disease Association (ABDA) Annual Scientific Meeting

Strategising Multidisciplinary Management of Breast Diseases

ORGANOTROPISM IN METASTATIC BREAST CANCER: A COMPARATIVE ANALYSIS OF BONE, BRAIN, AND LUNG INVOLVEMENT IN A TERTIARY REFERRAL CENTER



Lina Choridah[1], Antoinette D.I van Asselt[2], Ajeng Viska Icanervilia[1]

[1]Department Radiology, Faculty of Medicine, Public Health, and Nursing, Universitas Gadjah Mada, Yogyakarta, Indonesia •
2Department of Epidemiology, University Medical Centre Groningen, Groningen, The Netherlands

BACKGROUND

Organotropism describes the preferential spread of cancer cells to specific organs. Understanding this patterns in breast cancer may improve disease characterization and inform surveillance strategies

OBJECTIVES

This study evaluated **organ-specific metastatic involvement and radiological phenotypes** in patients with **breast cancer**

METHODS

DESIGN

Retrospective study conducted using medical records from a tertiarry referral center

SUBJECTS

A total of 1014 patients with metastases originating from five primary cancers treated between July and December 2024 at Dr. Sardjito General Hospital Yogyakarta.

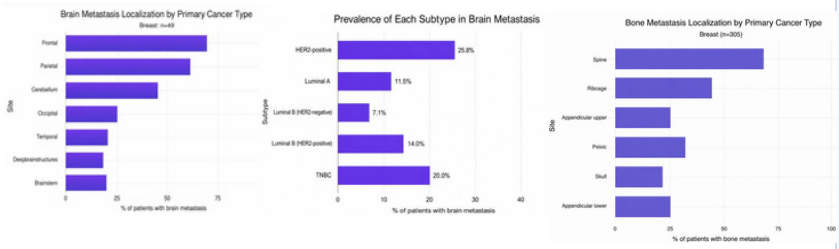
DATA

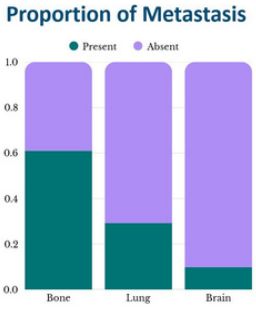
Patient characteristics, site-specific metastatic frequencies, anatomical distributions, radiological phenotypes, and organ-specific metastases according to molecular subtype were evaluated.

ANALYSIS

Categorical associations between breast cancer and radiological metastatic patterns were analyzed using the chi-square test with standardized residual analysis and compared with colorectal cancer.

RESULTS AND DISCUSSION





Logistic regression of pulmonary metastasis with colorectal cancer as reference

Variables in the comparison							
Step	Variable	B	S.E.	Wald	df	Sig.	Exp(B)
1	Breast cancer	-0.682	0.170	16.188	1	< 0.001	0.505

Logistic regression breast cancer and type of lung metastasis

Step	Variable	B	S.E.	Wald	df	Sig.	Exp(B)
1	Breast cancer	1.249	0.321	15.163	1	< 0.001	3.487

Bone Metastases

- In a cohort of 500 patients with metastatic breast cancer, bone was the most frequent site of distant metastasis, affecting 305 patients (61.0%).
- Spine was being the predominant skeletal location.

Lung Metastases

- Lung metastases were observed in 146 patients (29.2%), ranking as the **second most common metastatic site**.
- **Compared to colorectal cancer**, breast cancer associated with **significantly lower odds** of developing lung metastases (OR=0.505, *p*-0.005).
- Among patients with pulmonary metastases, breast cancer was significantly more likely than other primary malignancies to present with a **pneumonic pattern (OR = 3.487, p < 0.001)**, which also tended to demonstrate **unilateral distribution**

Brain Metastases

- Most located in the **frontal and parietal lobes**
- **HER2-enriched** molecular subtype showed a higher prevalence of brain involvement than the other breast cancer subtypes.

CONCLUSION / TAKE-HOME MESSAGE

Organotropism of Breast Cancer Metastasis

Breast cancer exhibits distinct organotropism characterized by predominant bone involvement, specific pulmonary characteristic, and cerebral metastatic patterns

Heterogeneity of Metastatic Pattern

Differences in metastatic distribution highlight the biological heterogeneity of metastatic spread

Next Steps

These findings may support risk stratification, personalized imaging surveillance, and earlier detection of

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ACKNOWLEDGEMENTS

- The study worked with secondary analysis of existing medical records, no additional diagnostic interventions were carried out on patients involved in the dataset.
- The ethical approval to gain access and work with the dataset was granted by the medical and health research ethics committee (MHREC) from the Universitas Gadjah Mada (UGM) and Dr. Sardjito General Hospital (Ref No KE/FK/0346/EC/2025).